

Anay Apte

331-401-8123 | apte1@engineering.upenn.edu | linkedin.com/in/anay-apte | github.com/AnayApte

EDUCATION

University of Pennsylvania

Philadelphia, PA

B.S.E. in Computer Science, 3.6/4.0

Aug. 2025 – May 2028

- **Relevant Coursework:** Programming Languages & Techniques, Data Structures & Algorithms, Designing LLM Agents, Language Modeling From Scratch (Stanford CS336, self-study), Discrete Mathematics, Probability, Multivariate Calculus, Linear Algebra.

EXPERIENCE

Software Engineering Intern

June 2026 — Present

Up Cancer

Atlanta, GA

- Architected a single-table AWS DynamoDB schema for an undocumented CRM frontend, designing PK/SK structure (e.g., **CONTACT#uuid** with **FAMILY#** child item) and a **GSI** per entity type.
- Chose single-table design for horizontal scalability, letting **20+ planned features** reuse schema.

Project Lead

January 2026 – Present

Penn Spark

Philadelphia, PA

- Led a team of **4 devs** and **3 designers** on SmartSketch, presenting at Demo Day to **100+ attendees**.
- Coordinated weekly sprints and cross-team syncs, aligning engineering and design throughout the revision process.

Software Engineering Intern

May 2024 – June 2025

Eigenvector LLC

Naperville, IL

- Built an AWS-backed VM deployment system with a **60-minute session limit**, used at **SkillsUSA Nationals**.
- Built a full-stack auth system using SvelteKit and PostgreSQL, verifying sign-ins via Cloudflare Turnstile.

PROJECTS

Scout | *Python, FastAPI, OpenAI API, Tavily API, SQLite*

- Built a system sourcing, screening, and issuing real investor decisions on founders within **24 hours**.
- Implemented a slide parser and GitHub/arXiv outbound scanner feeding into a multi-axis scorer with verification.
- Designed a shared contract across **4 independently-built modules** to avoid integration drift when deployed.

Language Modeling from Scratch | *Python, PyTorch*

- Implemented LLM components from scratch, including full Transformer architecture and a BPE tokenizer.

Penn Academic Co-Pilot | *Python, FastAPI, OpenAI API, pytest*

- Built an advising agent for planning schedules, checking course eligibility and degree progress via **18 GPT tools**.
- Built a conflict resolver across all lecture/recitation combinations, guarded by a layer blocking prompt-injection.

SmartSketch | *TypeScript, Next.js, React, Tailwind CSS, Supabase, Livekit, Gemini API, Deepgram*

- Built an app converting live speech into a structured, incrementally-updating mindmap to help ADHD notetaking.
- Built a pipeline where a Python agent joins a LiveKit room to stream audio into Gemini for mindmap generation.
- Regenerated the mindmap every **5 seconds/200 characters**, trading compute for guaranteed correctness.

ACTIVITIES

Director of Marketing & Singer

September 2025 – Present

Penn Masala (World's First South Asian A Cappella Group)

Philadelphia, PA

- Member of the world's first South Asian a cappella group with **400K+ followers** and content exceeding **10M+ views**, with group credits including performances at the **White House** and **2024 Paris Olympics**.

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, SQL (Postgres/SQLite), JavaScript, HTML/CSS, OCaml

Developer Tools: React, React Native, Next.js, Express, SvelteKit, FastAPI, AWS, DynamoDB, Git

Libraries: PyTorch, OpenAI API, Gemini API, Tavily API, Stripe API, Playwright, Pandas, NumPy